ALGORITHM 6: Optimal-DPOR algorithm for general blocking.

Initial call: $Explore(\langle\rangle, \emptyset, \langle\{\langle\rangle\}, \emptyset\rangle)$

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1  Explore(E, Sleep, WuT)
2      if enabled(s_[E]) = ∅ then
3          foreach e, e' ∈ dom(E) such that (e ≲_E e') do
4              let E' = pre(E, e);
5              let v = (notdep(e, E).ê');
6              if sleep(E') ∩ WI_[E'](v) = ∅ then
7                  wut(E') := insert_[E'](v, wut(E'));
8          foreach e ∈ dom(E) do
9              let E' = pre(E, e);
10             let w = notdep(e, E);
11             foreach subsequence u of w such that u ⊑_[E'] w do
12                 foreach p such that ê disables p after E'.u do
13                     let v = u.p;
14                     if sleep(E') ∩ WI_[E'](v) = ∅ then
15                         wut(E') := insert_[E'](v, wut(E'));
16     else
17         if WuT ≠ ⟨{⟨⟩}, ∅⟩ then
18             wut(E) := WuT;
19         else
20             choose p ∈ enabled(s_[E]);
21             wut(E) := ⟨{p}, ∅⟩;
22         sleep(E) := Sleep;
23         while ∃p ∈ wut(E) do
24             let p = min_≺{p ∈ wut(E)};
25             let Sleep' = {q ∈ sleep(E) | E ⊢ p◇q};
26             let WuT' = subtree(wut(E), p);
27             Explore(E.p, Sleep', WuT');
28             add p to sleep(E);
29             remove all sequences of form p.w from wut(E);
```

The proof of soundness for Algorithm 5 can be obtained by a small modification of the proof for Algorithm 1. As for Algorithm 3, we must replace w_R by w_q in Claim 5.4, Claim 5.5, and in related properties, just as in the proof for Algorithm 3. We define w'_q precisely as in the proof for Algorithm 3, and note that $w'_q \sqsubseteq_{[E.q]} w'$. Let w''_q be the reordering of the events in w'_q as they occur in w'. Then we can take u as w''_q at line 6 of the algorithm. By construction q disables $\widehat{e_q}$ after $E.w''_q$, and the rest of the proof proceeds as in the proof for Algorithm 3.

The extension of Algorithm 4 is shown in Algorithm 6. The corresponding extension of the race detection phase is now at lines 8–15.

Let us consider how to extend the soundness proof of Algorithm 2 to cover Algorithm 6. As for the locking case, the structure of the proof remains the same. The main modifications are the following: